

CE 310 — Week 3 Weekly Quiz

Week: 3 | **Topic:** Probability Basics: Marginal, Conditional, Joint, and Independence

Due: Closes **8:00 AM Tuesday, Sep 15** — the day before Exercise 3 and Homework 3 are due. Taken on D2L

Points: 15 questions $\times$ 1 pt each = 15 pts

Attempts: Two — the higher score counts

Time limit: 25 minutes

Instructions: Answer each question in the format it asks for — most are multiple choice, with one matching, one ordering, one select-all-that-apply and one true/false. This quiz is auto-graded and must be submitted before it closes at the time given above. You may use your notes and readings, but work independently.

This quiz covers the Monday lecture and the Wednesday Excel tutorial — marginal, conditional and joint probability, independence, Bayes' theorem and the base rate, and the COUNTIFS/AVERAGEIFS patterns that compute them in Excel. The tutorial's warning boxes are where the most time-consuming mistakes are flagged, so they are worth reading before you start.

Questions

Q1. An engineer wants to know whether high cooling demand is equally likely year-round. She calculates $P(\text{Cooling} > 50 \text{ kWh}) = 14.21\%$ across all 8,760 hours. Then she calculates $P(\text{Cooling} > 50 \text{ kWh} \mid \text{July}) = 38.31\%$. What does this difference reveal?

- A) Cooling demand is conditionally dependent on month — pooling all months hides the elevated seasonal risk.
 - B) The original 14.21% estimate contains a calculation error that must be corrected before use.
 - C) July data behaves as an outlier and should be excluded from the annual probability estimate used for design
 - D) Marginal and conditional probabilities must always agree when the sample is large enough.
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Q2. Before the 1986 Challenger launch, engineers had recorded O-ring failure data from previous flights. The critical mistake in the pre-launch analysis was:

- A) Using too small a sample — fewer than ten prior flights had been analysed before the final launch decision was made

- B) Reporting probabilities as decimals instead of percentages, causing a miscommunication.
 - C) Analyzing failures without conditioning on launch temperature, obscuring the temperature-failure relationship.
 - D) Confusing joint probability with marginal probability when combining failure modes from two boosters.
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Q3. A design standard states: “Size the cooling system for peak-hour demand.” Which probability concept best justifies conditioning on summer months and business hours when estimating that peak demand?

- A) The law of large numbers — larger samples converge to the true population mean regardless of conditioning.
 - B) The addition rule — $P(A \text{ or } B)$ accounts for overlap between summer and business-hours subsets.
 - C) Independence — summer hours and business hours are independent events, so their probabilities multiply.
 - D) Conditional probability — demand given the operating context matters more than the unconditional average.
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Q4. [True/False] $P(A | B)$ and $P(B | A)$ describe the same quantity, so once you have computed one you also know the other.

Q5. July is represented in the dataset by exactly 744 hourly rows. Which calculation correctly gives the marginal probability that a randomly selected hour falls in July?

- A) $744 / 365$
 - B) $744 / 12$
 - C) $8,760 / 744$
 - D) $744 / 8,760$
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Q6. A sensor flags a pump as faulty. When a fault is genuinely present the flag fires 90% of the time, and it also fires on 5% of healthy pumps. At any moment only 2% of pumps are faulty. Without computing an exact figure, what does Bayes’ theorem tell you about $P(\text{Fault} | \text{Flag})$?

- A) It is about 90%, because that is how often the flag is right when a fault is present
- B) It is well below 90% — false alarms outnumber true ones when faults are rare
- C) It is exactly 2%, because that is the base rate of faults

- D) It cannot be determined without knowing how many pumps were tested

Q7. [Select all that apply] Two events A and B are statistically independent. Which of the following must be true?

- (A) $P(A | B) = P(A)$
- (B) $P(B | A) = P(B)$
- (C) $P(A \text{ and } B) = P(A) \times P(B)$
- (D) $P(A \text{ and } B) = 0$
- (E) $P(A \text{ or } B) = P(A) + P(B)$

Q8. A building fault-detection system uses a Naive Bayes classifier to flag hours when equipment may be malfunctioning. The classifier estimates $P(\text{Electricity} > 150 \text{ kWh} | \text{Fault})$ and $P(\text{Electricity} > 150 \text{ kWh} | \text{Normal})$ from historical data, then applies Bayes' theorem to calculate the probability of a fault given the observed reading. Which statement best describes what the classifier is doing?

- A) Computing marginal probabilities and comparing them to a fixed threshold set by the equipment manufacturer.
- B) Using conditional probability — the likelihood of a high reading given each class — to update the fault probability.
- C) Checking statistical independence between electricity consumption and fault status before issuing an alert.
- D) Applying the addition rule to combine fault probabilities from multiple sensors into a single score.

Q9. [Matching] Match each Excel function to what it computes.

Choices:

- (A) `=COUNTIF(range, criterion)`
- (B) `=COUNTIFS(range1, crit1, range2, crit2)`
- (C) `=AVERAGEIF(range, criterion, average_range)`
- (D) `=AVERAGEIFS(average_range, range1, crit1, range2, crit2)`
- (E) `=COUNTA(range)`

Matches:

1. The mean of one column over the rows meeting a single condition
2. How many rows meet a single condition
3. The mean of one column over the rows meeting two conditions at once
4. How many rows meet two conditions at once

Q10. [Ordering] You want $P(\text{Cooling} > 50 \text{ kWh} \mid \text{July})$ from a table of 8,760 hourly rows. Put the steps in the order you would carry them out.

- (A) Restrict attention to the rows satisfying the condition — the 744 hours in July
 - (B) Count how many of those July hours have Cooling > 50 kWh
 - (C) Divide that count by 744
 - (D) Compare the result against the unconditional $P(\text{Cooling} > 50 \text{ kWh})$ to see whether conditioning changed the picture
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Q11. You use text criteria in COUNTIFS, counting rows where the Season column reads Summer. You get 0 and you were expecting hundreds. What is the most likely cause?

- A) Text criteria are case-sensitive, so Summer and summer are counted separately
 - B) A misspelled criterion returns 0 with no error at all, so nothing announces it
 - C) COUNTIFS cannot take text criteria at all; only COUNTIF will accept them
 - D) Text criteria must be entered without the surrounding quotation marks
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Q12. AVERAGEIF and AVERAGEIFS do not take their arguments in the same order. Which pair is right?

- A) AVERAGEIF(test_range, criterion, average_range) and AVERAGEIFS(average_range, test_range1, criterion1, ...)
 - B) Both take the average range first
 - C) Both take the test range first
 - D) AVERAGEIF(average_range, test_range, criterion) and AVERAGEIFS(test_range, criterion, average_range)
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Q13. You want the number of hours that are in Summer **and** above 50 kWh of cooling, with Season in column E and Cooling_kWh in column H. Which formula returns it?

- A) =COUNTIFS(E2:E8761, "Summer" AND H2:H8761, ">50")
 - B) =COUNTIF(E2:E8761, "Summer") * COUNTIF(H2:H8761, ">50")
 - C) =COUNTIFS(E2:E8761, "Summer", H2:H8761, ">50")
 - D) =COUNTIFS(E2:E8761 & H2:H8761, "Summer", ">50")
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Q14. You put =IF(H2>50, 1, 0) in cell M2 and fill it down to M8761. What does column M then contain?

- A) A 1 or a 0 for each row, according to that row's own Cooling_kWh value
 - B) The running count of rows so far whose cooling exceeds 50 kWh
 - C) A single 1 in M2 if any row in column H exceeds 50, and 0s elsewhere
 - D) The number of rows in column H that exceed 50 kWh, repeated down the column
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Q15. A conditional probability $P(A | B)$ is computed in Excel as a ratio of two counts. Which ratio?

- A) rows satisfying A and B, divided by rows satisfying B
- B) rows satisfying A and B, divided by the total number of rows
- C) rows satisfying B, divided by rows satisfying A
- D) rows satisfying A, divided by rows satisfying B